

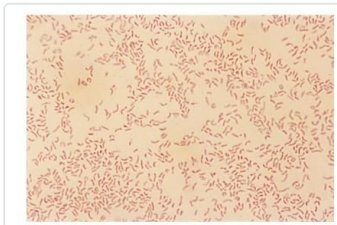
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What are the two toxigenic serogroups of *V. cholerae* that cause epidemic disease, and what distinguishes a biotype from a serotype?
2. What is the mechanism by which cholera toxin causes massive watery diarrhea?
3. Which culture medium is classically used to isolate *V. cholerae*, and what confirms it on dark-field microscopy?

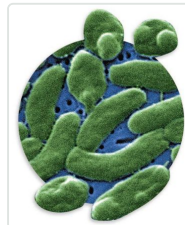
*Guessing wrong before reading still improves retention — attempt all three, then check as you go.*

**1 Vibrio cholerae — general features**

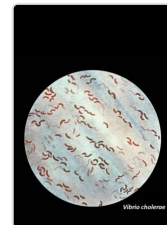
- **Curved (comma-shaped) rods** with a **single polar flagellum**; family **Vibrionaceae**, closely related to Enterobacteriaceae
- **Gram-negative**, motile, **oxidase-positive**, facultatively anaerobic
- **Grows well in alkaline conditions**; ferments on ordinary or selective media containing bile at 37°C
- Toxigenic strains **colonize the small intestine** and produce **cholera toxin**
- Classified into serogroups by its **O cell-wall antigen**



Comma-shaped bacillus with polar flagellum.



Gram stain of *Vibrio* spp. — curved and straight Gram-negative bacilli.



High-power view of curved *V. cholerae* cells.

**2 Serogroups and classification**

| Feature                     | Detail                                                                                     |
|-----------------------------|--------------------------------------------------------------------------------------------|
| <b>O-antigen serogroups</b> | <b>&gt;200 described</b> ; only toxigenic <b>O1</b> and <b>O139</b> cause epidemic cholera |
| <b>Non-O1 strains</b>       | Cause <b>sporadic disease</b> or are <b>nonpathogenic</b>                                  |
| <b>O1 biotypes</b>          | <b>Classic</b> and <b>El Tor</b> (El Tor = current global pandemic)                        |
| <b>O1 serotypes</b>         | <b>Ogawa, Inaba, Hikojima</b>                                                              |
| <b>O139</b>                 | Largely restricted to <b>Asia</b> , reported much less frequently now                      |

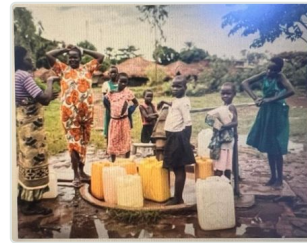
Biotype = differences in biochemical reactions; serotype = differences in antigenic structure — both used to characterize isolates in epidemiologic investigations.

**3 Transmission and epidemiology**

- Season/climate matter: **cold, acidic, dry** conditions inhibit *Vibrio*; **warm, monsoon, alkaline, saline** conditions favor it; the organism survives in water for long periods
- **Infective dose**:  $\sim 10^{10}$  organisms via water (acid-susceptible) vs  $10^2$ – $10^4$  via food (buffered by food matrix)
- Cholera has caused **seven historical pandemics**
- **2010 Haiti outbreak**: cholera reintroduced after nearly a century, unintentionally, by relief workers following the earthquake — **>770,000 infections,  $\geq 10,000$  deaths**



Flooded street — contaminated water is a key transmission route.



Populations without safe water access are at highest risk.

Rita Colwell — 50+ years studying *V. cholerae* and safe water in Bangladesh: found that cholera bacteria attach to plankton (copepods); filtering water through folded sari cloth removed 99% of bacteria and cut cholera cases by 50% ("her sari solution").



Rita Colwell.



Sari-cloth water filtration.

#### 4 Who is at higher risk

- No reliable access to **safe water, sanitation, and hygiene**
- Populations affected by **displacement, disaster, conflict, flooding**, or crowded living conditions
- **Travelers, aid workers, healthcare workers** in areas of active transmission
- **Reduced gastric acidity** increases susceptibility

#### 5 Clinical presentation

- Incubation: **hours to a few days**; onset is **abrupt vomiting**, followed by **copious watery feces**
- **"Rice-water" stool**: pale, watery, with flecks of mucus; **no RBCs or WBCs** in stool (non-invasive, secretory diarrhea)
- **No abdominal pain**; subsequent symptoms are all referable to **marked dehydration**
- Fluid/electrolyte loss → **cardiac and renal failure**; loss of bicarbonate and potassium → **acidosis and hypokalemia**
- Thirst and painful **muscle cramps** reflect volume/electrolyte loss
- Severe cases are **often afebrile**; **blood in stool suggests an alternative diagnosis**
- **Untreated mortality ≈ 40%**

Key teaching point: the danger is not the *number* of stools — it is the *speed and magnitude* of fluid loss.

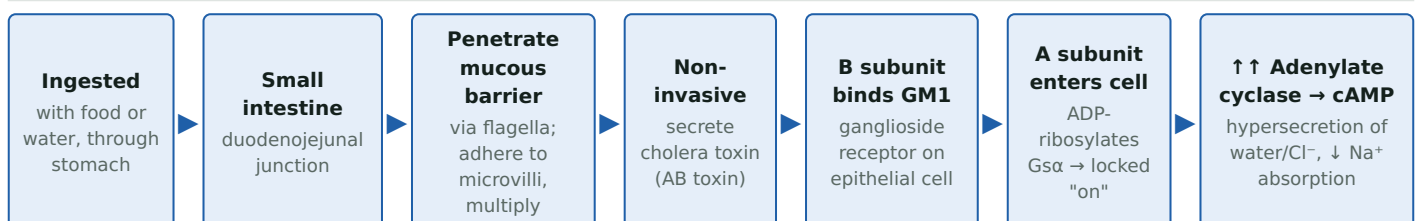


Rice-water stool sample.

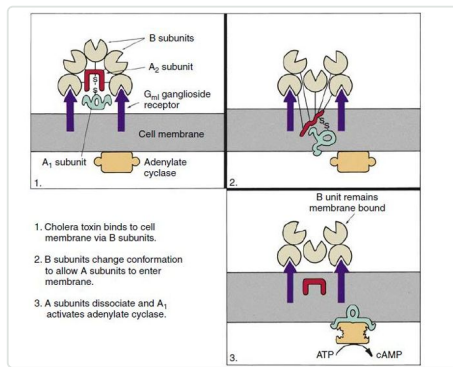


"Cholera cots" — designed to manage profuse diarrheal losses during outbreaks.

#### 6 Pathogenesis and virulence factors



Net effect: massive fluid and electrolyte loss → cramps, vomiting, fever, diarrhea. Quorum sensing regulates virulence-factor expression — a potential drug target.

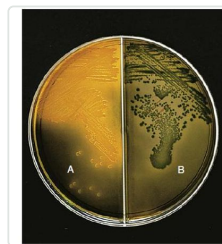


Structure and action of cholera toxin: B subunits bind GM1 → conformational change allows A subunit entry → A<sub>1</sub> activates adenylate cyclase → ATP → cAMP.

## 7 Laboratory diagnosis

- During an **epidemic**, diagnosis is largely **clinical** — little need for the lab
- **Dark-field microscopy**: curved cells with **brisk, darting motility** — confirmatory
- **Oxidase-positive** — distinguishes it from Enterobacteriaceae
- Group-specific antisera supportive

| Medium/Test           | Finding                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------|
| <b>MacConkey agar</b> | <b>Colorless colonies</b> (lactose fermented slowly)                                        |
| <b>TCBS agar</b>      | <b>Selective medium</b> widely used for isolation of <i>V. cholerae</i>                     |
| <b>TSI agar</b>       | <b>Acid slant / acid butt, no gas, no H<sub>2</sub>S</b> (ferments sucrose)                 |
| <b>Real-time PCR</b>  | Rapid testing of food/water; also detects <i>V. vulnificus</i> , <i>V. parahaemolyticus</i> |



TCBS agar: *V. cholerae* (A) vs *V. parahaemolyticus* (B) colony morphology.

## 8 Treatment and prevention

|                           | Detail                                                                                                                                                   |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mainstay</b>           | <b>Prompt fluid/electrolyte replacement</b> (oral or IV); <b>glucose added</b> to enhance water/electrolyte uptake                                       |
| <b>Antibiotics</b>        | <b>Adjunctive only</b> — for severe cases/high-risk patients, not routine for mild disease; ↓ diarrhea duration, stool volume, bacterial shedding        |
| <b>Antibiotic options</b> | <b>Doxycycline, azithromycin</b> (per local susceptibility)                                                                                              |
| <b>Pediatric care</b>     | <b>Zinc</b> for children 6 months–5 years when available; watch for <b>hypoglycemia</b> , dehydration, ongoing losses                                    |
| <b>Vaccine</b>            | <b>Vaxchora</b> — oral <b>live attenuated</b> vaccine (US, for travelers to O1-endemic areas); other <b>killed-organism</b> oral vaccines used elsewhere |
| <b>Chemoprophylaxis</b>   | <b>Tetracycline</b> effective for close contacts, but does <b>not</b> prevent spread of a major epidemic                                                 |
| <b>Public health</b>      | Clean water/food supply; <b>prompt carrier detection</b>                                                                                                 |

## 9 Non-cholera *Vibrio* species ("vibrioses")

- In the US, non-cholera *Vibrio* infection is **more common** than *V. cholerae* infection
- Main species: ***V. vulnificus*** and ***V. parahaemolyticus***
- Exposure via **seawater**, more often via **contaminated shellfish** (raw oysters)
- Rare but **increasing** — driven by ↑ raw oyster consumption, ↑ awareness/diagnosis, and a **warming climate**
- Can be **fatal in immunocompromised patients**, especially *V. vulnificus*; both are **nationally notifiable diseases** alongside *V. cholerae*

## P Must-remember pearls

- **Only O1 and O139** (toxigenic) cause epidemic cholera; O1 has **Classic** and **El Tor** biotypes and **Ogawa/Inaba/Hikojima** serotypes.
- Infective dose: **~10<sup>10</sup> via water**, but only **10<sup>2</sup>-10<sup>4</sup> via food** (food buffers stomach acid).
- Pathogenesis: cholera toxin B subunit binds **GM1**, A subunit **ADP-ribosylates Gsα** → constant adenylate cyclase activation → ↑cAMP → secretory diarrhea.
- Classic picture: **afebrile**, painless, profuse **"rice-water" stool**, **no fecal WBCs/RBCs** — danger is fluid-loss speed, not stool count.
- Diagnosis: **TCBS** selective agar, **dark-field motility**, **oxidase-positive**; treatment = **rehydration** first, antibiotics adjunctive (doxycycline/azithromycin); **Vaxchora** for travelers.

## Q Test yourself

**1. A 27-year-old man from a cholera-affected area develops sudden profuse watery diarrhea and vomiting. He is afebrile, thirsty, tachycardic and hypotensive; stool is pale and watery with mucus flecks. What is the most urgent management step?**

- |                                                        |                                        |
|--------------------------------------------------------|----------------------------------------|
| A. Prompt oral or IV fluid and electrolyte replacement | B. Empiric ciprofloxacin therapy       |
| C. Stool culture before any treatment                  | D. Abdominal CT to exclude perforation |

**2. Which two serogroups of V. cholerae are responsible for epidemic cholera?**

- |                  |                                |
|------------------|--------------------------------|
| A. O1 and O139   | B. O1 and O157                 |
| C. O139 and O111 | D. All >200 serogroups equally |

**3. What is the mechanism of action of cholera toxin?**

- |                                                                                                          |                                                                    |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| A. B subunit binds GM1; A subunit ADP-ribosylates Gsα, locking adenylate cyclase active and raising cAMP | B. It directly lyses enterocytes causing hemorrhagic colitis       |
| C. It inhibits protein synthesis at the 60S ribosomal subunit                                            | D. It forms pores in the epithelial membrane causing osmotic lysis |

**4. Which medium is classically used to selectively isolate V. cholerae from stool?**

- |               |                         |
|---------------|-------------------------|
| A. TCBS agar  | B. MacConkey agar alone |
| C. Blood agar | D. Sabouraud agar       |

**5. Which feature distinguishes V. cholerae stool findings from a typical invasive bacterial diarrhea?**

- |                                                             |                                          |
|-------------------------------------------------------------|------------------------------------------|
| A. Absence of fecal RBCs and WBCs despite large volume loss | B. Presence of numerous fecal leukocytes |
| C. Frank blood mixed uniformly in stool                     | D. Foul-smelling, greasy, floating stool |

### ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Rehydration is the single most important, life-saving intervention in cholera; antibiotics are only adjunctive.
- 2 — A** — Only toxigenic O1 and O139 strains cause widespread epidemic disease; other serogroups are sporadic or nonpathogenic.
- 3 — A** — Cholera toxin is an AB toxin; GM1 is the receptor, and constant Gsα activation drives cAMP overproduction and secretory diarrhea.
- 4 — A** — Thiosulphate-citrate-bile salts-sucrose (TCBS) agar is the selective medium widely used for isolation.
- 5 — A** — Cholera is a non-invasive, secretory (toxin-mediated) diarrhea — rice-water stool contains no RBCs or WBCs.

### SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam